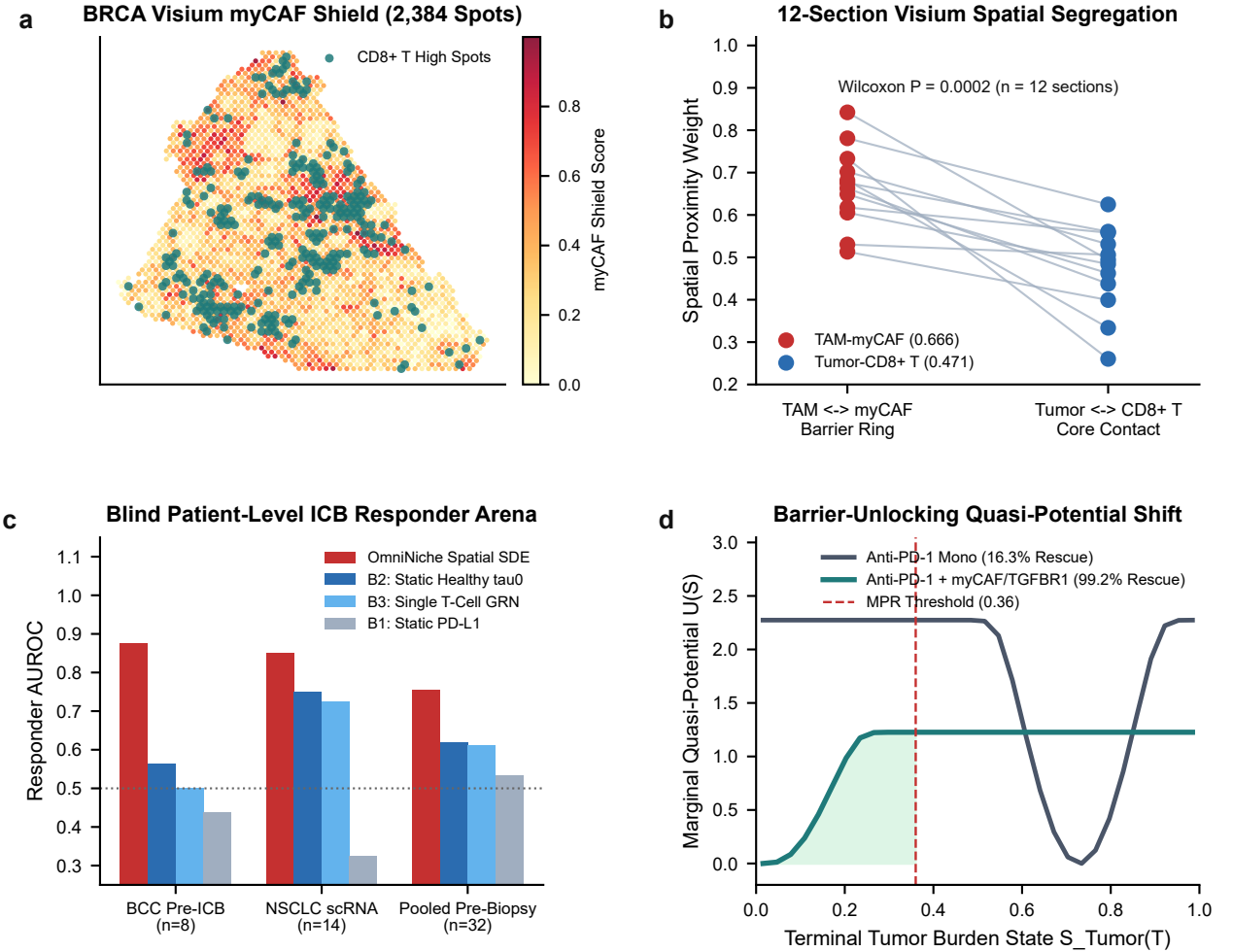

a
BRCA Visium myCAF Shield (2,384 Spots)
CD8+ T High Spots
myCAF Shield Score
0.0
0.2
0.4
0.6
0.8
b
12-Section Visium Spatial Segregation
Wilcoxon P = 0.0002 (n = 12 sections)
Spatial Proximity Weight
TAM-myCAF (0.666)
Tumor-CD8+ T (0.471)
TAM <-> myCAF
Barrier Ring
Tumor <-> CD8+ T
Core Contact
c
Blind Patient-Level ICB Responder Arena
OmniNiche Spatial SDE
B2: Static Healthy tau0
B3: Single T-Cell GRN
B1: Static PD-L1
Responder AUROC
BCC Pre-ICB
(n=8)
NSCLC scRNA
(n=14)
Pooled Pre-Biopsy
(n=32)
d
Barrier-Unlocking Quasi-Potential Shift
Anti-PD-1 Mono (16.3% Rescue)
Anti-PD-1 + myCAF/TGFBR1 (99.2% Rescue)
MPR Threshold (0.36)
Marginal Quasi-Potential U(S)
Terminal Tumor Burden State S_Tumor(T)